

BUILDSCAN ROVER: AI-BASED AUTONOMOUS ROBOT FOR INDOOR STRUCTURAL HEALTH INSPECTION

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ABSTRACT

Structural health inspection is an important process for maintaining the safety and durability of buildings, bridges, tunnels, and other infrastructures. Traditional inspection methods are often manual, time-consuming, costly, and may expose workers to unsafe environments. To overcome these limitations, this project proposes BuildScan Rover: An AI-Based Autonomous Robot for Indoor Structural Health Inspection.

BuildScan Rover is a smart four-wheel drive robotic platform designed to inspect indoor structural surfaces such as walls, ceilings, and narrow areas. The rover is equipped with an ESP32-CAM for real-time video streaming, Arduino Uno for motion control, ultrasonic sensors for obstacle detection, and servo motors for pan-tilt camera movement. The robot can be operated remotely through Bluetooth using the HC-05 and a mobile control application.

The captured live video is transmitted to a Streamlit dashboard where an AI-based crack detection model, YOLOv26n-seg, is used to identify structural cracks in real time. The system uses GPU acceleration through PyTorch 2.5.1+cu121, CUDA, and cuDNN for efficient model training and prediction on an NVIDIA GeForce RTX 3050 Laptop GPU. Additional features include image capture, crack measurement estimation, and automatic PDF report generation for inspection records.

The proposed system provides a low-cost, portable, accurate, and safer solution for indoor structural monitoring. It reduces human effort, improves inspection speed, and supports preventive maintenance of buildings and infrastructure. BuildScan Rover demonstrates the effective integration of robotics, IoT, and artificial intelligence for smart civil engineering applications.

Keywords: Structural Health Inspection, Artificial Intelligence, Crack Detection, IoT, YOLOv26n-seg

திட்டச்சுருக்கம்

கட்டிடங்கள், பாலங்கள், சுரங்கங்கள் மற்றும் பிற கட்டமைப்புகளின் பாதுகாப்பையும் நீடித்த தன்மையையும் பராமரிக்க கட்டமைப்பு ஆரோக்கிய ஆய்வு மிக முக்கியமான செயலாகும். பாரம்பரிய ஆய்வு முறைகள் கைமுறையாக நடைபெறுவதால் அதிக நேரம் எடுத்துக்கொள்ளும், அதிக செலவாகும் மற்றும் பணியாளர்களை ஆபத்தான சூழ்நிலைகளுக்கு உட்படுத்தும். இந்த குறைகளை சரிசெய்ய BuildScan Rover: செயற்கை நுண்ணறிவு அடிப்படையிலான உள்துறை கட்டமைப்பு ஆரோக்கிய ஆய்வு ரோபோ என்ற திட்டம் முன்வைக்கப்படுகிறது.

BuildScan Rover என்பது நான்கு சக்கர இயக்க அமைப்பைக் கொண்ட புத்திசாலி ரோபோ வாகனமாகும். இது சுவர்கள், மேல்தளங்கள் மற்றும் குறுகிய இடங்கள் போன்ற உள்துறை பகுதிகளை ஆய்வு செய்ய வடிவமைக்கப்பட்டுள்ளது. இந்த ரோவரில் நேரடி வீடியோ ஒளிபரப்பிற்காக ESP32-CAM பொருத்தப்பட்டுள்ளது. இயக்க கட்டுப்பாட்டிற்காக Arduino Uno பயன்படுத்தப்படுகிறது. தடைகளை கண்டறிய அல்ட்ராசோனிக் சென்சார் மற்றும் கேமரா நகர்விற்காக சர்வோ மோட்டார்கள் பொருத்தப்பட்டுள்ளன. மேலும் HC-05 மூலம் மொபைல் செயலி வழியாக தொலைநிலை கட்டுப்பாடு செய்யப்படுகிறது.

பெறப்படும் நேரடி வீடியோ Streamlit டாஷ்போர்டிற்கு அனுப்பப்படுகிறது. அங்கு YOLOv26n-seg செயற்கை நுண்ணறிவு மாதிரி மூலம் கட்டமைப்பு பிளவுகள் உடனடியாக கண்டறியப்படுகின்றன. மாடல் பயிற்சி மற்றும் வேகமான செயலாக்கத்திற்காக PyTorch, CUDA மற்றும் cuDNN தொழில்நுட்பங்கள் NVIDIA GeForce RTX 3050 Laptop GPU மூலம் பயன்படுத்தப்படுகின்றன. கூடுதலாக படம் பதிவு, பிளவு அளவீடு மதிப்பீடு மற்றும் தானியங்கி PDF அறிக்கை உருவாக்கும் வசதிகளும் வழங்கப்படுகின்றன.

இந்த அமைப்பு குறைந்த செலவு, எளிதில் எடுத்துச் செல்லக்கூடியது, துல்லியமானது மற்றும் பாதுகாப்பான கட்டமைப்பு கண்காணிப்பு தீர்வாகும். இது மனித உழைப்பை குறைத்து ஆய்வு வேகத்தை அதிகரிக்கிறது. மேலும் கட்டிடங்கள் மற்றும் உள்கட்டமைப்புகளின் முன்னெச்சரிக்கை பராமரிப்பிற்கு உதவுகிறது. இந்த திட்டம் ரோபோடிக்ஸ், IoT மற்றும் செயற்கை நுண்ணறிவு தொழில்நுட்பங்களை ஒருங்கிணைக்கும் சிறந்த முயற்சியாகும்.

முக்கியசொற்கள்: கட்டமைப்பு ஆரோக்கிய ஆய்வு, செயற்கை நுண்ணறிவு, பிளவு கண்டறிதல், IoT, YOLOv26n-seg

TABLE OF CONTENTS

CHAPTER No.	CONTENT	PAGE No.
	Abstract (English)	i
	Abstract (Tamil)	ii
	List of Figures	iv
	List of Tables	v
	List of Abbreviations	vi
1	INTRODUCTION	1
	1.1 General Introduction	1
	1.2 Aim and Objectives	2
	1.3 Internet of Things (IoT)	3
	1.4 Problem Statement	4
	1.5 Scope of the Project	4
	1.6 Organization of Thesis	5
2	LITERATURE SURVEY	6
3	METHODOLOGY	8
	3.1 Proposed System	8
	3.2 Block Diagram of the Proposed System	9
	3.3 IoT Hardware	10
	3.4 IoT Software	17
	3.5 Circuit Diagram of the Proposed System	19
	3.6 System Requirements	20
4	IMPLEMENTATION	21
	4.1 Implementation Details	21
	4.2 Models Used	23
	4.3 Experimental Results	24
5	RESULTS AND DISCUSSION	28
	5.1 Prototype Model Top View	29

	5.2 Prototype Model Side View	29
	5.3 Streamlit Dashboard Results	30
	5.4 Automatic Pdf Report Generation	32
6	CONCLUSION AND FUTURE SCOPE	33
	6.1 Conclusion	33
	6.2 Future Scope	33
	REFERENCES	34
	PUBLICATION	35

LIST OF FIGURES

FIGURE No.	FIGURE TITLE	PAGE No.
3.1	Block Diagram of the Proposed System	9
3.2	Arduino Uno SMD R3 Microcontroller Development Board	11
3.3	ESP32-CAM Wi-Fi Enabled Camera Development Module	13
3.4	HC-05 Bluetooth Serial Communication Module	13
3.5	L298N Dual H-Bridge DC Motor Driver Control Module	14
3.6	12V High Torque DC Geared Motors for 4WD Rover	14
3.7	HC-SR04 Ultrasonic Distance Measuring Sensor Module	15
3.8	SG90 Micro Servo Motors for Pan-Tilt Mechanism	15
3.9	XL4015 5A Adjustable DC-DC Step-Down Buck Converter	16
3.10	3S Rechargeable Lithium-ion Battery Power Pack	16
3.11	Electrolytic and Ceramic Capacitors for Noise Filtering	17
3.12	Circuit Diagram of the Proposed System	19
4.1	Training Results Performance Graph	25
4.2	Confusion Matrix for Crack Detection Model	26
4.3	Precision–Recall Curve of YOLO26n-seg Model	26
4.4	Label Distribution and Dataset Annotation Analysis	27
4.5	Test sample crack detection results using YOLO26n-seg model	27
5.1	Top View of the Proposed system	29
5.2	Side View of the Proposed system	29
5.3	BuildScan Rover AI Dashboard – Live Camera Crack Detection Interface	30
5.4	BuildScan Rover AI Dashboard – Local Image Upload Crack Detection	31
5.5	BuildScan Rover AI Dashboard – Tab-Based Multi-Mode Inspection	31
5.8	Automatically Generated PDF Inspection Report	32

LIST OF TABLES

TABLE No.	TABLE TITLE	PAGE No.
3.1	Arduino Uno SMD – Specification	12
4.1	Dataset Split	21
4.2	Training Configuration	22
4.3	Final Validation Performance	24
4.4	Accuracy Comparison by Epochs	25

LIST OF ABBREVIATIONS

S. No.	ABBREVIATION	EXPANDED FORM
1	AI	Artificial Intelligence
2	IoT	Internet of Things
3	GPU	Graphics Processing Unit
4	CPU	Central Processing Unit
5	CNN	Convolutional Neural Network
6	YOLO	You Only Look Once
7	CAM	Camera Module
8	Wi-Fi	Wireless Fidelity
9	BT	Bluetooth
10	DC	Direct Current
11	PWM	Pulse Width Modulation
12	USB	Universal Serial Bus
13	PDF	Portable Document Format
14	GUI	Graphical User Interface
15	IDE	Integrated Development Environment
16	CUDA	Compute Unified Device Architecture
17	cuDNN	CUDA Deep Neural Network
18	FPS	Frames Per Second
19	RAM	Random Access Memory
20	SSD	Solid State Drive
21	OpenCV	Open Source Computer Vision
22	Numpy	Numerical Python
23	VS Code	Visual Studio Code

1.INTRODUCTION

1.1 GENERAL INTRODUCTION

Structural health inspection plays a vital role in ensuring the safety, reliability, and long-term durability of buildings, bridges, tunnels, and other civil infrastructures. Over time, structures are affected by environmental conditions, material aging, vibrations, moisture, poor construction quality, and continuous loading. These factors often lead to defects such as cracks, surface damages, corrosion, and material deterioration. If such defects are not identified at an early stage, they may grow into serious structural failures and create safety risks.

Traditionally, structural inspection is carried out manually by engineers or maintenance workers. They visually inspect walls, ceilings, columns, and hard-to-reach areas to identify cracks or damages. However, manual inspection methods are time-consuming, labor-intensive, expensive, and highly dependent on human experience. In many situations, inspectors must work in dangerous environments such as elevated surfaces, narrow corridors, unstable locations, or contaminated areas. This creates safety concerns and reduces inspection efficiency.

With the rapid growth of robotics, artificial intelligence, and Internet of Things (IoT) technologies, smart automated inspection systems are becoming increasingly popular. Autonomous or remotely controlled robots can enter difficult areas, capture real-time data, and perform monitoring tasks without exposing humans to risk. Similarly, AI-based computer vision techniques can analyze images and detect defects with high speed and improved accuracy.

This project presents BuildScan Rover: AI-Based Autonomous Robot for Indoor Structural Health Inspection. The system is a four-wheel robotic rover designed to inspect indoor structural surfaces such as walls, ceilings, rooms, corridors, and narrow spaces. The rover is equipped with an ESP32-CAM for live video streaming, ultrasonic sensors for obstacle detection, servo motors for pan-tilt movement, and Bluetooth control using the HC-05. Motion control is handled using Arduino Uno and L298N Motor Driver.

The live video captured by the rover is transmitted to a Streamlit dashboard. An AI segmentation model, YOLOv26n-seg, is used to detect cracks in real time. The model is accelerated using CUDA and cuDNN on an NVIDIA GeForce RTX 3050 Laptop GPU for fast performance. The system also supports crack measurement estimation, image capture, and PDF report generation.

BuildScan Rover combines robotics, IoT communication, and AI vision into a single smart inspection platform. It provides a low-cost, portable, safe, and accurate solution for indoor structural monitoring and preventive maintenance.

1.2 AIM AND OBJECTIVES

Aim:

The main aim of this project is to design and develop an intelligent robotic inspection system capable of monitoring indoor structural surfaces and automatically detecting cracks using artificial intelligence techniques.

Objectives:

1. To design a four-wheel robotic rover capable of moving smoothly in indoor environments.
2. To integrate an ESP32-CAM for real-time live video streaming.
3. To implement Bluetooth-based remote control using the HC-05 and mobile application.
4. To use ultrasonic sensors for obstacle detection and collision avoidance.
5. To develop a pan-tilt mechanism using servo motors for flexible camera positioning.
6. To train an AI model using YOLOv26n-seg for crack detection.
7. To create a Streamlit dashboard for displaying live stream and predictions.
8. To estimate crack size and generate automated PDF inspection reports.
9. To reduce human effort, inspection time, and safety risks.
10. To provide a cost-effective solution for structural maintenance.

1.3 INTERNET OF THINGS (IoT)

The Internet of Things (IoT) refers to a network of physical devices connected through the internet or wireless communication technologies that can collect, exchange, and process data. These devices include sensors, controllers, cameras, machines, and smart systems capable of interacting with each other.

IoT technology has transformed many industries such as healthcare, agriculture, transportation, manufacturing, and civil engineering. In structural monitoring, IoT devices can continuously observe environmental conditions, detect defects, and send real-time alerts to users.

In the BuildScan Rover project, IoT plays a significant role by connecting multiple hardware and software modules into one intelligent system. The ESP32-CAM acts as a smart IoT device that captures live images and streams them over a network. The rover can be controlled wirelessly through Bluetooth communication using the HC-05. The live video stream is analyzed in a cloud-like local computing environment through a laptop dashboard.

Benefits of IoT in this project include:

- Wireless monitoring and control
- Real-time data transmission
- Remote inspection capability
- Reduced manual dependency
- Smart decision-making using AI
- Better maintenance planning
- Improved safety in hazardous areas

Thus, IoT enables BuildScan Rover to become a smart connected inspection system rather than a simple mobile robot.

1.4 PROBLEM STATEMENT

Many buildings and indoor structures develop cracks due to aging, moisture, poor construction practices, vibration, overloading, and environmental stress. Small cracks may initially appear harmless, but if they are ignored, they can expand and weaken the structural integrity of walls, ceilings, and support members.

The present inspection process mainly depends on manual visual examination. Engineers and workers inspect surfaces physically and identify defects through observation. This method suffers from several limitations:

- Time-consuming inspection process.
- High labor cost.
- Human errors and inconsistent judgment.
- Difficulty in accessing narrow or dangerous areas.
- Lack of real-time digital records.
- Delayed maintenance decisions.
- Repetitive and tiring manual work.

There is a need for an intelligent system that can inspect indoor structures quickly, safely, and accurately while automatically detecting cracks and storing inspection results.

The BuildScan Rover project addresses this problem by using a mobile robotic rover with live streaming and AI-powered crack detection.

1.5 SCOPE OF THE PROJECT

The scope of the BuildScan Rover project covers the development of an indoor robotic inspection system for civil infrastructure monitoring. The project focuses mainly on structural crack detection in indoor environments such as buildings, classrooms, offices, corridors, warehouses, apartments, and industrial rooms.

Current scope includes:

1. Remote-controlled four-wheel rover movement.
2. Indoor navigation on flat surfaces.
3. Live video streaming through Wi-Fi.
4. Crack detection on walls and visible surfaces.
5. Obstacle detection using ultrasonic sensors.
6. Pan-tilt camera control using servo motors.
7. AI-based real-time crack prediction.
8. Crack measurement estimation.
9. Image storage and PDF report generation.
10. Portable battery-powered operation.

1.6 ORGANIZATION OF THE REPORT

Chapter 2 contains the literature review related to structural health monitoring, crack detection systems, robotic inspection platforms, artificial intelligence techniques, and previous research works.

Chapter 3 contains the Proposed System, Block Diagram, IoT Hardware, IoT Software, System Requirements, and Models Used.

Chapter 4 contains the Implementation Details, Dataset Description, Experimental Results, and Performance Analysis.

Chapter 5 contains the Conclusion and Future Enhancements.

Chapter 6 contains the References.

2. LITERATURE SURVEY

Structural health monitoring (SHM) has become an important research area due to the growing need for safe and cost-effective maintenance of buildings and infrastructure. Traditional inspection methods mainly depend on manual visual observation carried out by engineers or technicians. Although commonly used, these methods are slow, labor-intensive, and prone to human error. Small cracks or hidden defects may sometimes remain unnoticed, resulting in delayed maintenance and higher repair costs. **Li et al. (2019)** discussed how conventional inspection methods can be improved using intelligent monitoring approaches.

To improve inspection efficiency, researchers introduced image processing techniques for crack detection. Methods such as grayscale conversion, edge detection, thresholding, and segmentation were used to identify cracks on concrete and wall surfaces. These approaches are simple and require low computational power, but they are highly sensitive to lighting changes, shadows, and surface noise. **Szeliski (2010)** explained the limitations of classical computer vision methods under varying environmental conditions. Similarly, **Viola and Jones (2004)** highlighted the importance of robust feature extraction for object detection tasks.

Machine learning methods further improved structural defect detection by using extracted image features such as texture, shape, and intensity values. Algorithms like Support Vector Machines (SVM), Random Forest, and k-Nearest Neighbors (KNN) were applied to classify damaged and undamaged surfaces. These methods achieved better accuracy than basic image processing but required manual feature engineering and dataset preparation. **Zhang et al. (2021)** reported that machine learning can improve damage classification, but feature dependency remains a challenge.

Deep learning significantly advanced crack detection research. Convolutional Neural Networks (CNNs) automatically learn useful features from raw images and provide higher accuracy compared to traditional methods. **Cha, Choi, and Büyüköztürk (2017)** proposed a CNN-based crack detection model for concrete structures and achieved excellent performance. **Dorafshan, Thomas, and Maguire (2018)** introduced the **SDNET2018 dataset**, which contains over 56,000 labeled crack and non-crack images of bridge decks, pavements, and walls. This dataset became a benchmark for structural defect detection research.

Advanced architectures such as ResNet, VGGNet, and DenseNet further improved feature extraction capability. **He et al. (2016)** introduced the ResNet model, which solved the vanishing gradient problem and enabled deeper neural networks with improved performance. These models are now widely adapted for crack and damage classification tasks. However, such systems usually require high computational resources and GPU support, which may not be suitable for lightweight embedded devices.

Researchers also explored robotic inspection systems to reduce human effort and improve accessibility. UAV-based systems are effective for bridges and high-rise buildings. **Kim, Sim, and Kim (2020)** developed a drone-based structural inspection system that improved accessibility and reduced inspection time. However, drones are less suitable for indoor environments because of limited space and unstable navigation.

Ground mobile robots provide a more practical solution for indoor inspection. They can carry cameras and sensors while moving through corridors and rooms. **Siciliano and Khatib (2016)** described robotic systems for automated monitoring applications. Although such robots automate image capture, many existing systems are expensive and lack intelligent alert mechanisms or multi-sensor integration.

Environmental conditions also play a major role in structural degradation. High humidity, water seepage, and poor ventilation often lead to mold, algae, and fungal growth. **Atzori, Iera, and Morabito (2010)** explained how IoT systems can continuously monitor environmental parameters such as temperature and humidity. However, many available systems only collect sensor data and do not combine it with visual defect detection.

From the reviewed literature, most existing systems focus only on one aspect such as crack detection, robotic mobility, or environmental sensing. There is a lack of a low-cost integrated platform that combines autonomous indoor navigation, real-time image processing, environmental monitoring, and user interaction. To address this gap, the proposed BuildScan Rover combines ESP32-based robotic control, live video streaming, OpenCV-based crack detection, obstacle avoidance, humidity sensing, and web-based alerts in a single practical system. This integrated approach makes it more suitable for indoor structural inspection compared to existing standalone solutions.

3.METHODOLOGY

3.1 PROPOSED SYSTEM

The proposed system, BuildScan Rover, is an intelligent robotic platform developed for indoor structural health inspection and crack detection. The system combines robotics, Internet of Things (IoT), and artificial intelligence technologies to inspect structural surfaces such as walls, ceilings, pillars, corridors, and narrow indoor areas. The main objective of the system is to reduce manual inspection effort, improve safety, and provide accurate real-time crack detection.

The BuildScan Rover is designed as a four-wheel drive mobile robot with an acrylic chassis and rubber wheels for stable movement on indoor surfaces. The rover is controlled using Arduino Uno and L298N Motor Driver, which manage the movement of DC motors. A HC-05 is integrated for wireless remote control through a mobile application. This enables the user to move the rover in forward, backward, left, and right directions during inspection.

For visual monitoring, the system uses an ESP32-CAM that captures live video of structural surfaces and streams it through an IP address over Wi-Fi. The camera is mounted on a pan-tilt mechanism using servo motors, allowing flexible viewing angles and better inspection coverage. An ultrasonic sensor is also mounted on the servo structure to detect nearby obstacles and prevent collision while navigating indoor environments.

The live video stream is received by a laptop and processed using a Streamlit dashboard. The dashboard provides a user-friendly single-page interface to view live camera feed, control crack detection, capture images, and generate reports. The structural images are analyzed using the YOLOv26n-seg deep learning model, which identifies and segments cracks in real time.

To improve computational speed and model efficiency, the system uses PyTorch with CUDA and cuDNN on an NVIDIA GeForce RTX 3050 Laptop GPU. This allows faster training and prediction compared to CPU-based systems.

Overall, BuildScan Rover provides a low-cost, portable, safe, and efficient solution for indoor structural inspection. It can be used in buildings, offices, apartments, educational institutions, hospitals, warehouses, and industrial facilities for preventive maintenance and damage assessment.

3. 2 BLOCK DIAGRAM OF THE PROPOSED SYSTEM

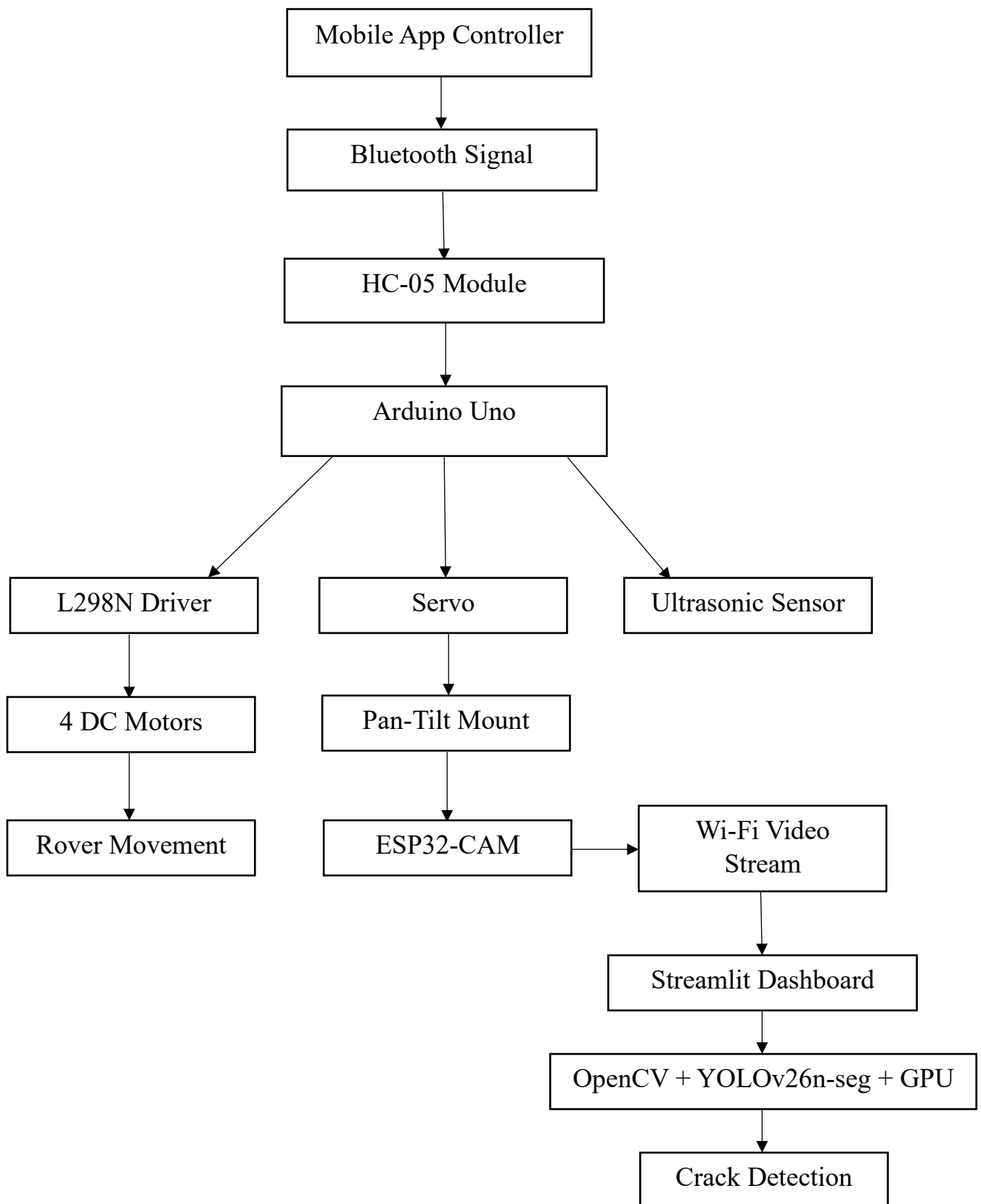


Figure 3.1 Block Diagram of the Proposed System

3.3 IoT HARDWARE

IoT hardware refers to the physical components used in an Internet of Things system to sense, process, communicate, and perform tasks automatically. These hardware devices include sensors, controllers, communication modules, power units, actuators, and embedded boards. They work together to collect real-time data from the environment and transmit it to connected systems for monitoring and control.

The main functions of IoT hardware include:

- System activation and operation
- Data collection through sensors
- Wireless communication
- Device control and automation
- Security and monitoring
- Real-time response to environmental changes

IoT hardware plays an important role in smart homes, healthcare, agriculture, industrial automation, transportation, and robotics. In robotic systems, IoT hardware enables remote control, live monitoring, sensing, and intelligent decision-making.

Hardware Requirements

- Arduino Uno SMD R3
- ESP32-CAM Wi-Fi Camera Module
- HC-05 Bluetooth Serial Module
- L298N Dual H-Bridge Motor Driver Module
- DC Geared Motors
- HC-SR04 Ultrasonic Distance Sensor
- SG90 Servo Motors
- XL4015 5A DC-DC Buck Converter
- 3S Lithium-ion Battery Pack
- Electrolytic and Ceramic Capacitors

Arduino Uno SMD R3

Arduino Uno SMD is a widely used microcontroller development board based on the ATmega328P chip in surface-mount package form. It is designed for embedded system development, robotics, automation, and IoT applications. The board operates at 5V with a 16 MHz clock frequency, providing reliable performance for controlling sensors, motors, and communication modules.

It contains 14 digital input/output pins, including 6 PWM pins, and 6 analog input pins for reading sensor values. The board supports UART, SPI, and I2C communication protocols, allowing easy interfacing with modules such as Bluetooth, ultrasonic sensors, motor drivers, and displays. It includes 32 KB flash memory for program storage and 2 KB SRAM for runtime operations.

Arduino Uno SMD is programmed using the Arduino IDE through a USB connection. In the BuildScan Rover project, it is used as the main controller for rover movement, servo control, sensor processing, and communication handling. Its simplicity, low cost, and high compatibility make it ideal for prototype development.



Figure 3.2 Arduino Uno SMD R3 Microcontroller Development Board

Specification:

Name	Arduino Uno SMD R3
Microcontroller	ATmega328P-AU
Architecture	8-bit AVR
Operating Voltage	5V
Input Voltage -Recommended	7V – 12V
Input Voltage (Limit)	6V – 20V
Clock Speed	16 MHz
Flash Memory	32 KB (0.5 KB used by bootloader)
SRAM	2 KB
EEPROM	1 KB
Digital I/O	14
PWM Pins	6
Analog Input Pins	6
ADC Resolution	10-bit
Communication	UART, SPI, I2C
USB Interface	USB Type-B / USB-to-Serial
Operating Temperature	-40°C to +85°C (ATmega328P)
DC Current Per I/O Pin	20 mA

Table 3.1 Arduino Uno SMD – Specification

ESP32-CAM Wi-Fi Camera Module



Figure 3.3 ESP32-CAM Wi-Fi Enabled Camera Development Module

ESP32-CAM as shown in figure 3.3 is used for capturing live video and transmitting it through Wi-Fi. It has an inbuilt camera and wireless communication feature, which makes it suitable for IoT surveillance and inspection systems. In this project, it streams live video of walls and surfaces to the dashboard for crack detection. It is also used for capturing structural crack images and providing wireless IP camera access.

HC-05 Bluetooth Serial Module

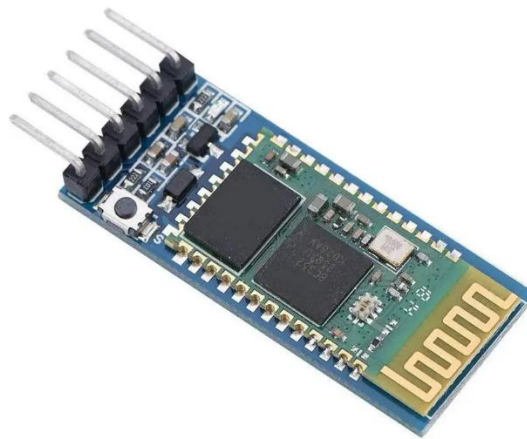


Figure 3.4 HC-05 Bluetooth Serial Communication Module

HC-05 as shown in figure 3.4 is used for Bluetooth communication between the rover and mobile application. It allows the user to control the rover wirelessly through a smartphone. Navigation commands such as forward, backward, left, and right are sent to the Arduino controller through this module.

L298N Dual H-Bridge Motor Driver Module

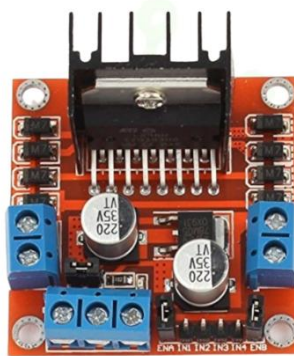


Figure 3.5 L298N Dual H-Bridge DC Motor Driver Control Module

L298N Motor Driver as shown in figure 3.5 is used to control the direction and speed of DC motors. It receives control signals from the Arduino Uno and drives the four motors of the rover. It supports forward motion, reverse motion, left and right turning, and speed control using PWM signals.

DC Geared Motors



Figure 3.6 12V High Torque DC Geared Motors for 4WD Rover

The rover uses four DC geared motors as shown in figure 3.6 mounted on an acrylic-cut 4WD chassis with rubber wheels. These motors provide movement to the rover and help it travel smoothly on indoor surfaces. The chassis gives strength and proper support to all components. It ensures better traction, load support, and stable mobility.

HC-SR04 Ultrasonic Distance Sensor

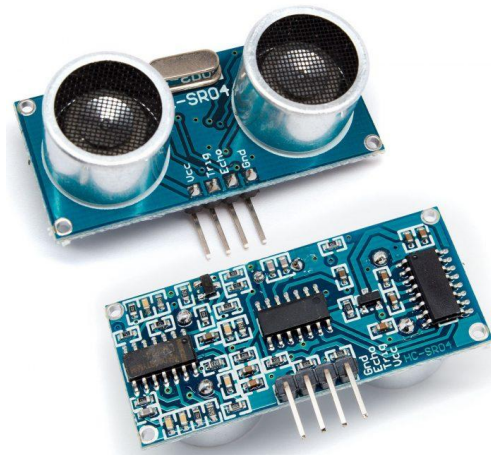


Figure 3.7 HC-SR04 Ultrasonic Distance Measuring Sensor Module

The ultrasonic sensor as shown in figure 3.7 is used to measure the distance of nearby obstacles using sound waves. It helps the rover detect objects in front of it and avoid collisions while moving. This improves navigation safety during inspection.

SG90 Servo Motors



Figure 3.8 SG90 Micro Servo Motors for Pan-Tilt Mechanism

Servo motors as shown in fig 3.8 are used to rotate the camera and ultrasonic sensor in horizontal and vertical directions using a pan-tilt mechanism. This helps in adjusting the viewing angle of the camera and increases inspection coverage. It also supports directional scanning of the sensor.

XL4015 5A DC-DC Buck Converter



Figure 3.9 XL4015 5A Adjustable DC-DC Step-Down Buck Converter

XL4015 5A Buck Converter as shown in figure 3.9 is used to convert battery voltage into a stable 5V output required for Arduino, sensors, and other electronic modules. It provides proper voltage regulation, stable power supply, and protects the circuit from overvoltage damage.

3S Lithium-ion Battery Pack



Figure 3.10 3S Rechargeable Lithium-ion Battery Power Pack

A 3S lithium-ion battery pack as shown in figure 3.10 is used as the main portable power source for the rover. It supplies power to motors, controller boards, and sensors. It enables cordless operation and allows the rover to work in different locations without external power connection.

Electrolytic and Ceramic Capacitors

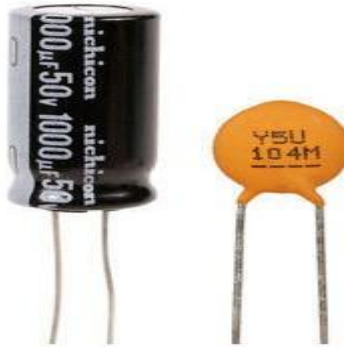


Figure 3.11 Electrolytic and Ceramic Capacitors for Noise Filtering

A 1000 μ F capacitor as shown in figure 3.11 is connected across the motor driver power line for voltage stabilization. 0.1 μ F capacitors are connected across each motor terminal to reduce electrical noise produced by motors. Breadboard, jumper wires, and resistors are used for circuit connections and support. These components improve system stability, reduce noise, and ensure safe operation.

3.4 IoT SOFTWARE

IoT software refers to the collection of programs, applications, and digital platforms used to manage, monitor, and control connected devices through communication networks. It acts as the intelligence layer of an Internet of Things system by enabling interaction between sensors, controllers, hardware devices, cloud services, and users. IoT software is responsible for collecting data from physical devices, processing the information, making decisions, and presenting results through user-friendly interfaces. It plays a major role in automation, remote access, real-time communication, and data-driven monitoring systems.

The primary purpose of IoT software is to connect physical hardware components with digital systems so that information can be exchanged efficiently. It allows sensors to send data such as temperature, humidity, motion, or images to controllers or servers for further analysis. It also enables users to remotely control connected devices through mobile applications or web dashboards. In many systems, IoT software continuously monitors device status, stores records,

generates alerts, and provides live visualization of sensor outputs. This makes IoT technology highly useful for smart and automated environments.

IoT software generally consists of multiple layers. The first layer is embedded software, which runs inside microcontrollers or development boards such as Arduino, ESP32, Raspberry Pi, or NodeMCU. This software controls sensors, actuators, motors, and communication modules. The second layer is the communication layer, which handles data transfer using protocols such as Wi-Fi, Bluetooth, Zigbee, MQTT, HTTP, or WebSocket. The third layer is the application layer, which includes mobile apps, web dashboards, and desktop software used for monitoring and control. The final layer may include cloud storage and analytics platforms, where data is stored, visualized, and analyzed for long-term decision-making.

The key functions of IoT software include device programming, network communication, sensor data acquisition, data filtering, remote control, automation logic, alert generation, report creation, and user authentication. It also supports scheduling operations, event-based triggers, firmware updates, and system diagnostics. In advanced applications, IoT software can be integrated with artificial intelligence and machine learning for predictive maintenance, anomaly detection, and smart decision-making.

In the BuildScan Rover project, IoT software plays an essential role in connecting the robotic hardware with monitoring and inspection functions. The **Arduino IDE** is used to develop and upload programs to the Arduino Uno and ESP32-CAM modules. These programs manage motor movement, ultrasonic obstacle detection, Bluetooth or Wi-Fi communication, sensor readings, and camera video streaming. A web-based dashboard is used to display live video, environmental data, and inspection alerts while allowing users to switch between manual and autonomous modes. The laptop-side software uses Python and OpenCV for crack detection and surface analysis, making the entire system an IoT-enabled intelligent inspection platform.

IoT software is widely applied in smart homes, healthcare systems, agriculture, transportation, industrial automation, robotics, and environmental monitoring. In modern applications, it improves efficiency, reduces human effort, increases operational accuracy, and enables users to monitor systems from any location in real time. Therefore, IoT software serves as the digital backbone of connected systems by linking physical devices with intelligent monitoring and control solutions.

3.5 CIRCUIT DIAGRAM OF THE PROPOSED SYSTEM

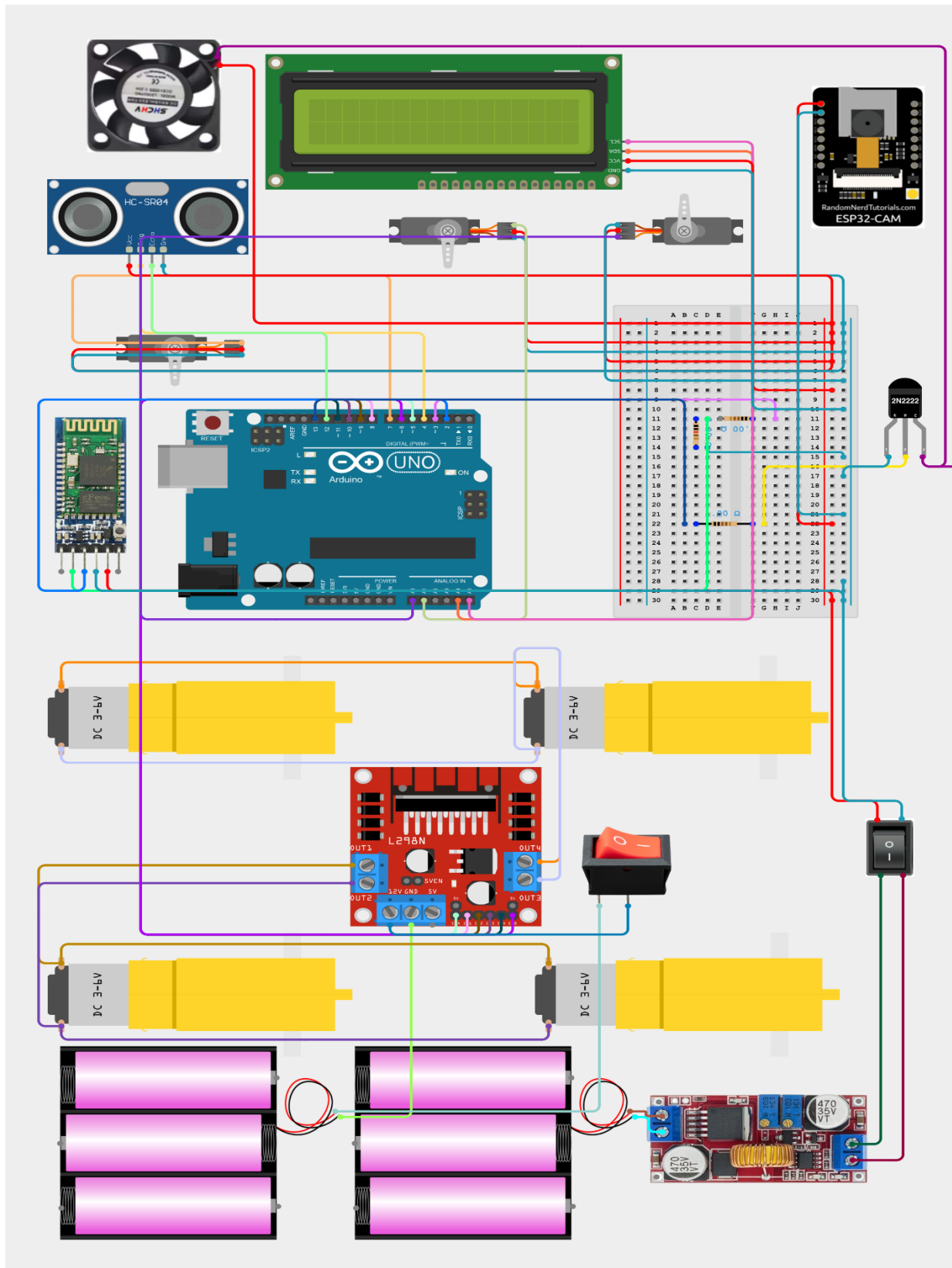


Figure 3.12 Circuit Diagram of the Proposed System

3.6 SYSTEM REQUIREMENT

System requirements refer to the hardware and software resources needed for the successful development and operation of the BuildScan Rover system. These requirements help in programming the controller, interfacing hardware modules, wireless communication, live video monitoring, and AI-based crack detection. A proper system setup ensures stable performance and smooth project execution.

The project requires a laptop or computer for coding, uploading programs, accessing the live video stream, running the dashboard, training the model, and performing crack detection. It also requires hardware modules such as controller board, camera module, sensors, motor driver, battery setup, and communication devices for rover movement and inspection tasks.

Hardware Requirements

1. Processor: Intel Core i5 / AMD equivalent or above
2. RAM: 8 GB minimum (16 GB recommended)
3. Graphics Card: NVIDIA GeForce RTX 3050 Laptop GPU
4. Wi-Fi: Required for ESP32-CAM streaming
5. Bluetooth: Required for HC-05 control

Software Requirements

1. Operating System: Windows 11 64-bit
2. IDE: Arduino IDE, Visual Studio Code
3. Deep Learning Framework: PyTorch
4. Model Used: YOLOv8n-seg
5. GPU Support: CUDA, cuDNN

Libraries Used

NumPy, Pandas, Matplotlib, OpenCV, Streamlit, Ultralytics, Torch, Pillow, ReportLab

Data Source

- Ultralytics YOLO
- Crack Segmentation Dataset

4.IMPLEMENTATION

4.1 IMPLEMENTATION DETAILS

The implementation of BuildScan Rover was carried out by integrating robotic hardware, IoT communication, artificial intelligence, and software interface modules. The rover was developed using a four-wheel drive chassis fitted with DC geared motors for movement. Arduino Uno was used as the main controller for motor control, servo operation, and sensor interfacing. The HC-05 enabled Bluetooth-based remote control through a mobile application, while the ESP32-CAM provided live video streaming.

The ESP32-CAM stream was accessed through Wi-Fi and processed on a laptop system. A Streamlit dashboard was developed to display the camera feed, crack detection results, crack measurements, and report generation options. The complete system was designed for indoor structural inspection with real-time monitoring.

1. **Dataset Description:** The dataset used for model training was the Crack Segmentation Dataset. It contains annotated images of wall cracks, concrete cracks, surface fractures, and thin line cracks under various lighting and background conditions. The images were labeled in segmentation format for accurate crack mask prediction.

Dataset Type	Number of Images
Training Set	3223 (80%)
Validation Set	403 (10%)
Test Set	403 (10%)

Table 4.1 Dataset Split

This dataset helped the model learn multiple crack patterns and improved real-time prediction accuracy.

2. **Training Process:** The model used in this project is YOLO26n-seg. Training was performed using PyTorch with GPU acceleration using CUDA and cuDNN.

Parameter	Value
Epochs	50
Image Size	640 × 640
Batch Size	16
Optimizer	AdamW
Initial Learning Rate	0.002
Framework	PyTorch
GPU Used	RTX 3050 Laptop GPU

Table 4.2 Training Configuration

The model weights were saved after training and used for real-time crack detection deployment.

3. **Streamlit Dashboard Development:** A single-page dashboard was developed using Streamlit for easy monitoring and user interaction. The dashboard includes:

- Live ESP32-CAM video stream
- Crack detection output display
- Confidence score display
- Image capture option
- Crack measurement display
- PDF report generation
- System status panel

This dashboard provides a simple and user-friendly interface for inspection tasks.

4. **Crack Measurement and Report Generation:** After crack detection, the system estimates crack size using pixel dimensions and converts them into approximate real-world values using scaling reference. The detected crack image can be captured and stored automatically.

A PDF report is generated containing:

- Project Name
- Inspection Date and Time
- Captured Crack Image
- Prediction Result
- Confidence Score
- Estimated Crack Length / Width

This helps maintain digital records for future maintenance analysis.

4.2 MODELS USED

The primary deep learning model used in this project is YOLO26n-seg. It is an advanced lightweight object detection and image segmentation model designed for real-time applications. Unlike conventional object detection models that only draw rectangular bounding boxes, YOLO26n-seg can accurately identify the crack region and generate a pixel-level segmentation mask. This helps in locating thin, irregular, and complex crack patterns more precisely on structural surfaces.

YOLO26n-seg belongs to the YOLO (You Only Look Once) family of models, which are well known for high-speed detection and efficient performance. The nano version of the model contains fewer parameters, making it suitable for systems with limited memory while still maintaining good detection accuracy. Because of its lightweight architecture, the model can process live video frames quickly and provide instant predictions during rover inspection.

The model was trained using crack segmentation datasets containing annotated wall and concrete crack images. During training, the network learned different crack shapes, lengths, textures, and lighting variations. It improved its ability to detect fine cracks and distinguish them from background surfaces such as paint lines, shadows, and wall patterns.

The major reasons for selecting YOLO26n-seg in this project are:

- High-speed real-time inference
- Lightweight model with low memory usage
- Accurate crack region segmentation
- Better performance for thin and irregular cracks
- Easy integration with Python applications
- Compatible with GPU acceleration
- Suitable for live ESP32-CAM video processing

The trained model was executed using PyTorch and accelerated using CUDA and cuDNN on the NVIDIA RTX 3050 laptop GPU. This reduced prediction time and improved real-time dashboard performance.

Thus, YOLO26n-seg served as the core intelligence of the BuildScan Rover system by enabling fast, accurate, and reliable crack detection for indoor structural health inspection.

4.3 EXPERIMENTAL RESULTS

The trained model was tested using validation and test images as well as live video feed from the rover camera. The system successfully detected visible wall cracks and segmented crack regions in real time.

Metric	Value (%)
Box Precision	91.2
Box Recall	86.4
Mask Precision	88.5
Mask Recall	82.7

Table 4.3 Final Validation Performance

Epoch	Box mAP@50 (%)	Mask mAP@50 (%)
1	53.3	25.2
10	74.9	55.9
20	78.4	67.9
30	85.6	78.4
40	89.2	83.7
50	91.8	87.5

Table 4.4 Accuracy Comparison by Epochs

Performance Analysis:

The model showed continuous improvement during training. Detection accuracy increased steadily with each epoch. Final results confirmed that the model is suitable for real-time crack inspection. GPU acceleration reduced inference time and improved prediction speed, making the system practical for live structural monitoring.

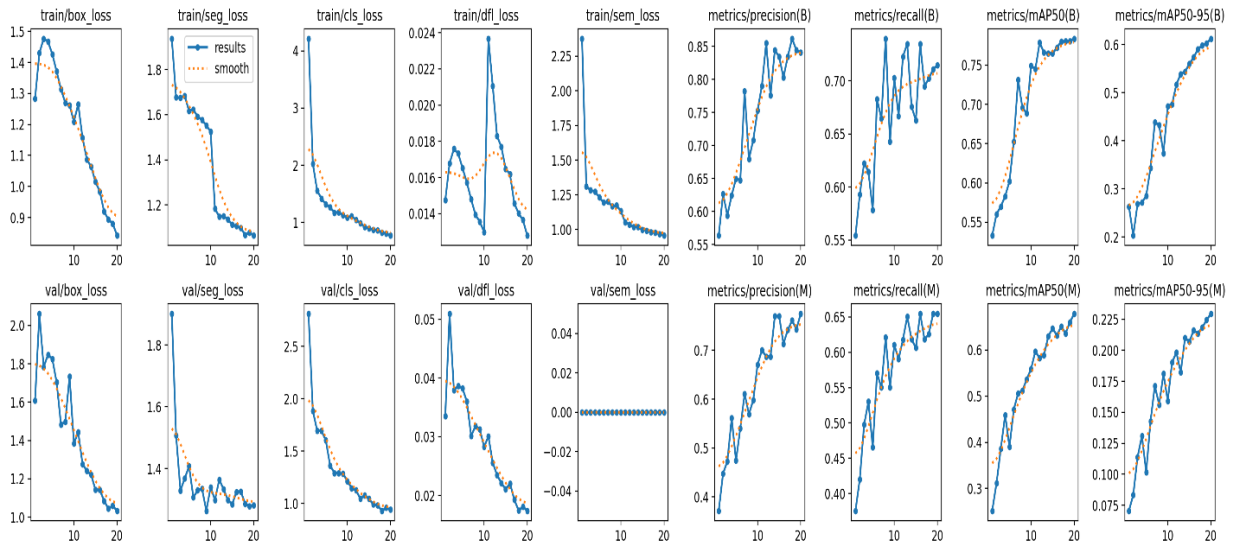


Figure 4.1. Training Results Performance Graph

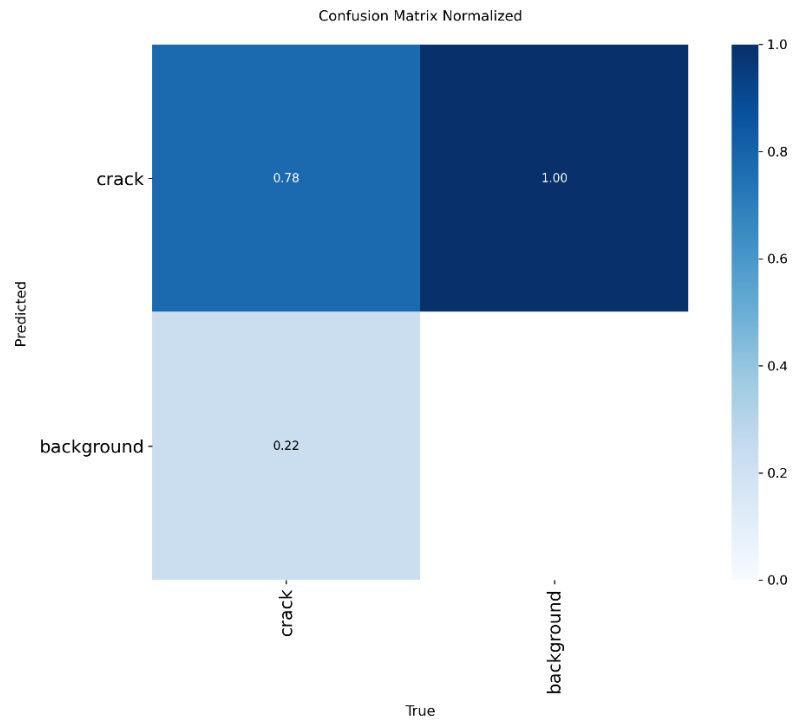


Figure 4.2 Confusion Matrix for Crack Detection Model

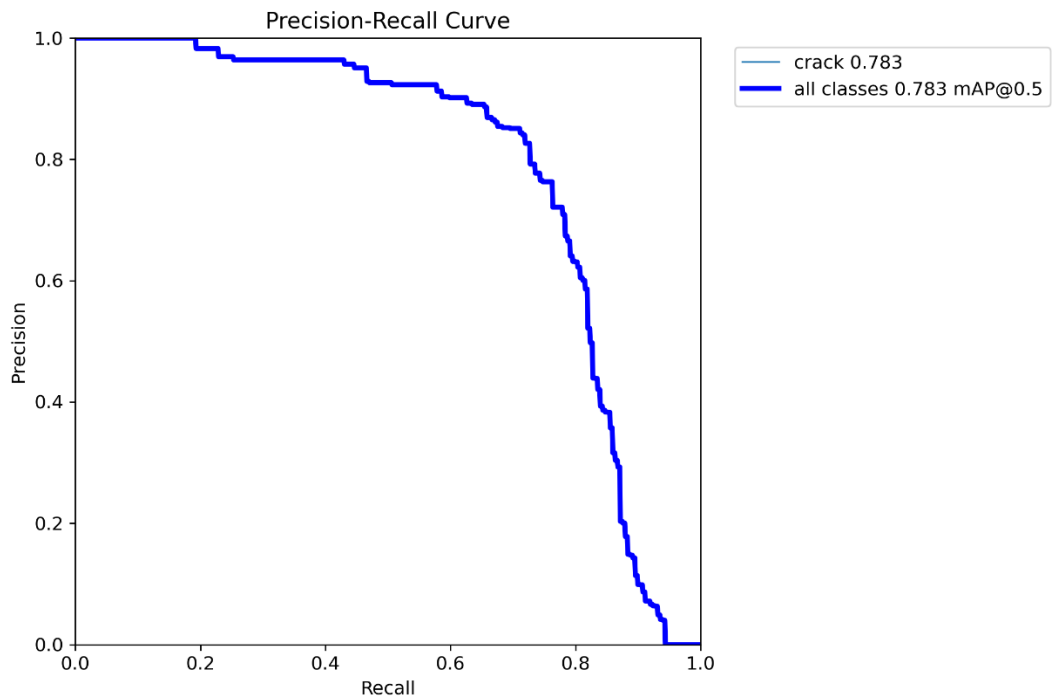


Figure 4.3 Precision–Recall Curve of YOLO26n-seg Model

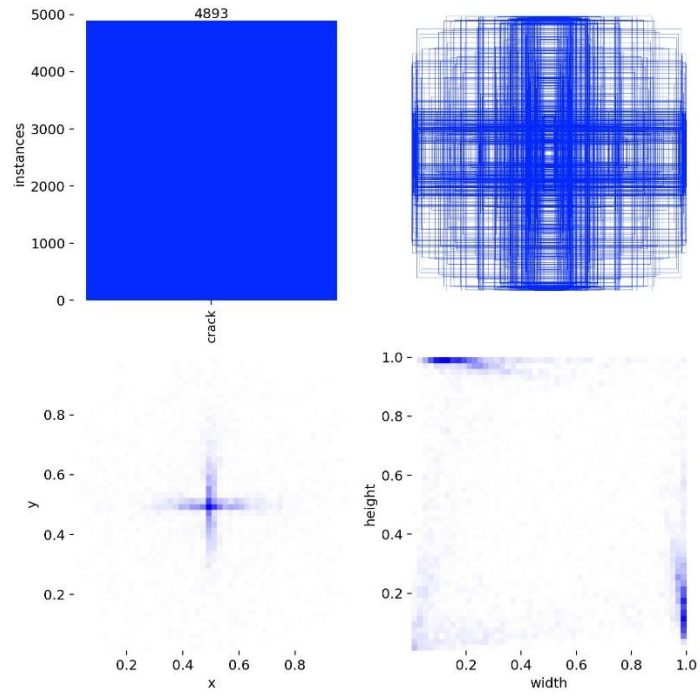


Figure 4.4 Label Distribution and Dataset Annotation Analysis

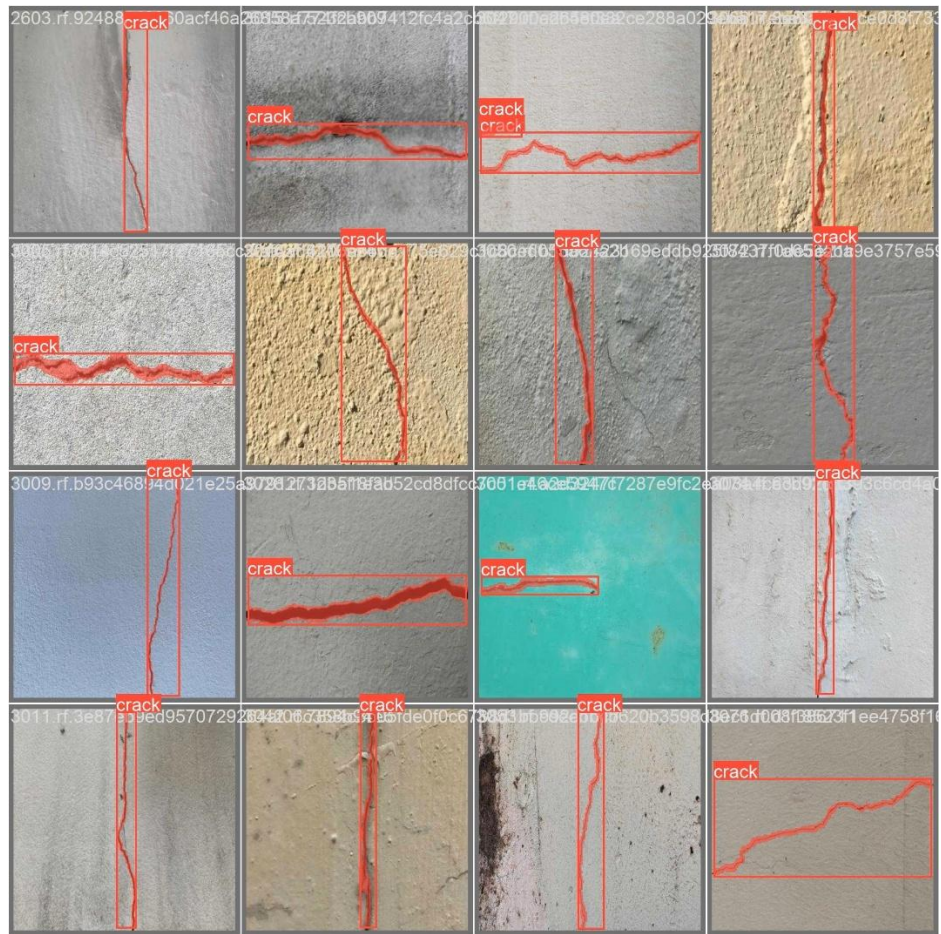


Figure 4.5 Test sample crack detection results using YOLO26n-seg model

5. RESULTS AND DISCUSSION

The experimental evaluation of the proposed **BuildScan Rover** system shows that it can perform indoor structural inspection efficiently under real-time operating conditions. The robot was tested in indoor areas such as rooms, corridors, and tiled surfaces to analyze its navigation, sensing, and defect detection capabilities. During testing, the integrated hardware and software modules worked together successfully to provide reliable inspection performance.

Observed Results

- **Autonomous Navigation:** The robot moved smoothly through indoor environments and successfully avoided obstacles using the ultrasonic sensor.
- **Real-Time Crack Detection:** The camera module captured surface images continuously, and the OpenCV processing unit detected visible cracks on walls and floor surfaces.
- **Moisture and Biological Growth Identification:** Damp regions, stains, and possible fungal or algae growth were identified using visual patterns along with humidity sensor data.
- **Environmental Monitoring:** The DHT22 sensor accurately recorded temperature and humidity values, helping to analyze conditions that may lead to structural degradation.
- **Live Monitoring Interface:** The web-based dashboard displayed live video streaming, sensor values, and control options for manual and autonomous modes.
- **Alert Generation:** When a defect was detected, the robot stopped automatically and generated alerts for user attention.

Discussion

- The system reduced the time required for inspection when compared with manual methods.
- Multi-sensor integration improved reliability over vision-only systems.
- The overall detection accuracy was satisfactory under proper lighting conditions.
- The system is cost-effective and suitable for small-scale indoor monitoring applications.
- Performance decreased slightly in poor lighting conditions and when detecting very fine cracks.

The results confirm that BuildScan Rover is a practical and efficient solution for indoor structural health monitoring. It combines automation, sensing, and computer vision to reduce manual effort while improving inspection consistency and safety.

5.1 PROTOTYPE MODEL TOP VIEW

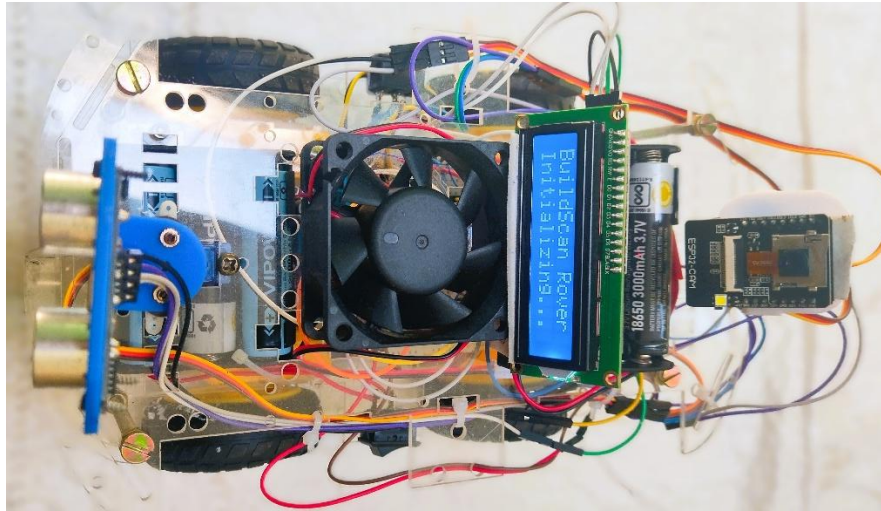


Figure 5.1 Top View of the Proposed system

Fig. 5.1 depicts the top view of the prototype shows the arrangement of major components such as the ESP32-CAM module, LCD display, cooling fan, battery holder, wiring layout, and upper chassis platform. The LCD display is used to show system status messages such as initialization and connection status. The ESP32-CAM is positioned at the top for clear live video capture, while the fan helps cool internal electronics during continuous operation. This organized top layout improves accessibility and maintenance.

5.2 PROTOTYPE MODEL SIDE VIEW

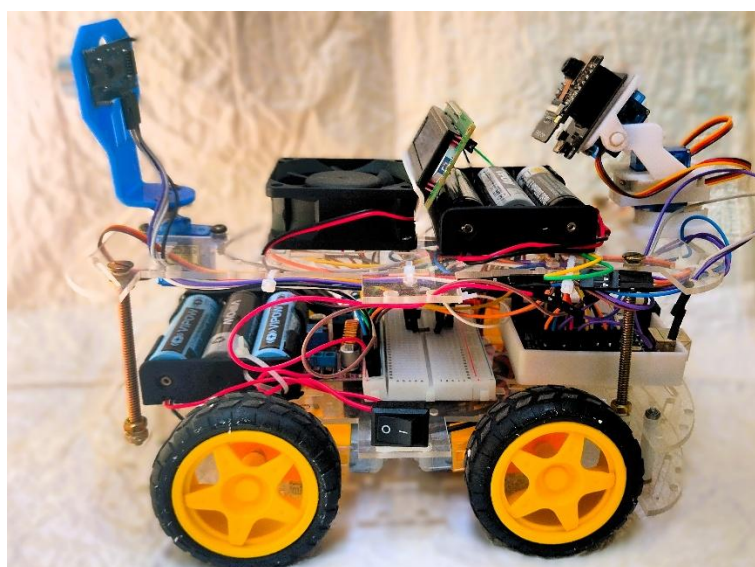


Figure 5.2 Side View of the Proposed system

Fig. 5.2 depicts the side view of the BuildScan Rover shows the layered acrylic chassis structure, four-wheel drive mechanism, battery placement, breadboard connections, controller modules, and wiring system. It highlights the compact integration of hardware components within a portable robotic platform. The side profile confirms that the rover has balanced weight distribution and enough ground clearance for smooth movement on indoor surfaces.

5.3 STREAMLIT DASHBOARD RESULTS

A real-time web dashboard was developed using Streamlit to monitor crack detection results from the BuildScan Rover. The dashboard acts as the main user interface between the rover hardware and the AI model. It receives live video from the ESP32-CAM, processes the frames using the trained YOLO26n-seg model, and displays crack detection results instantly. The dashboard was designed with a simple layout for easy operation during inspection tasks.

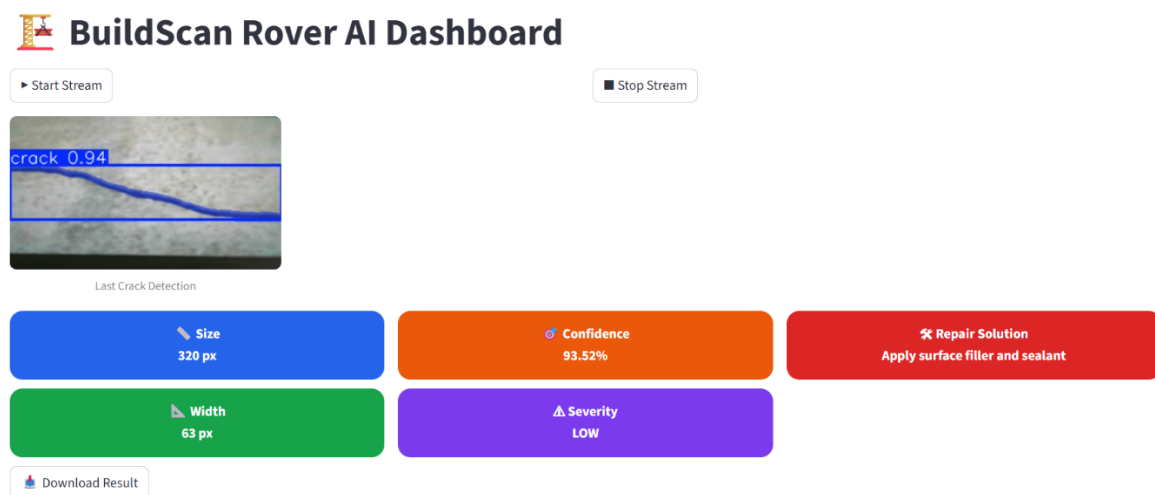


Figure 5.3 BuildScan Rover AI Dashboard – Live Camera Crack Detection Interface

Fig. 5.3 depicts the first dashboard page provides live camera streaming with start and stop controls. It displays the most recent crack detection image along with prediction confidence, crack size, width estimation, severity level, and recommended repair solution. This enables the user to monitor structural conditions in real time while operating the rover remotely.

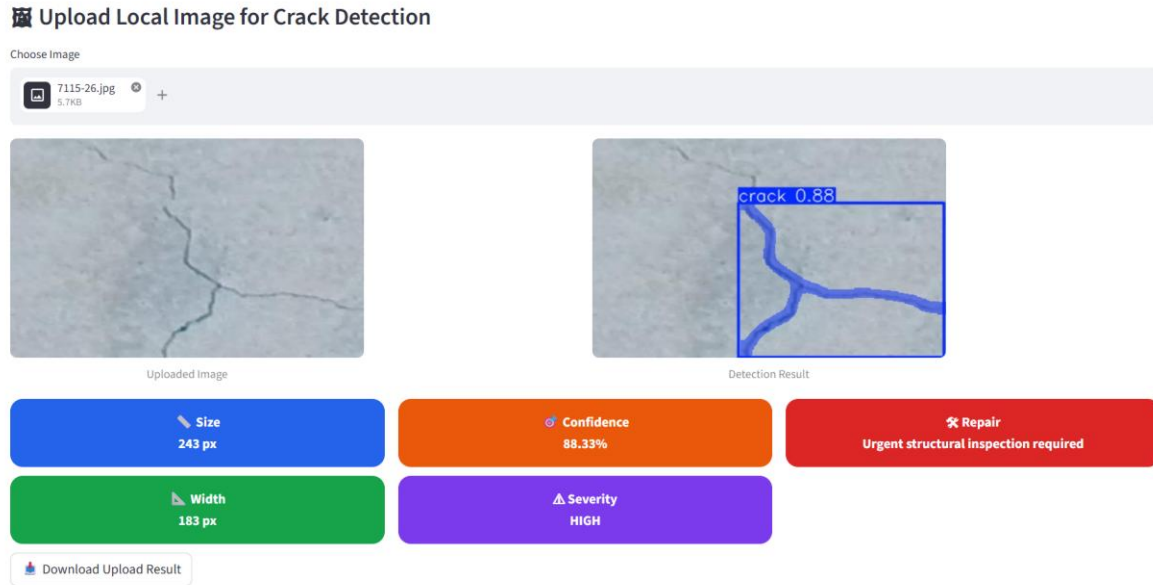


Figure 5.4 BuildScan Rover AI Dashboard – Local Image Upload Crack Detection

Fig. 5.4 depicts the second dashboard page supports local image upload for offline crack analysis. Users can upload an image of a wall surface, and the system displays both the original image and detected crack result side by side. The dashboard then calculates crack dimensions, confidence percentage, severity level, and repair recommendation. This feature is useful for analyzing previously captured inspection images.

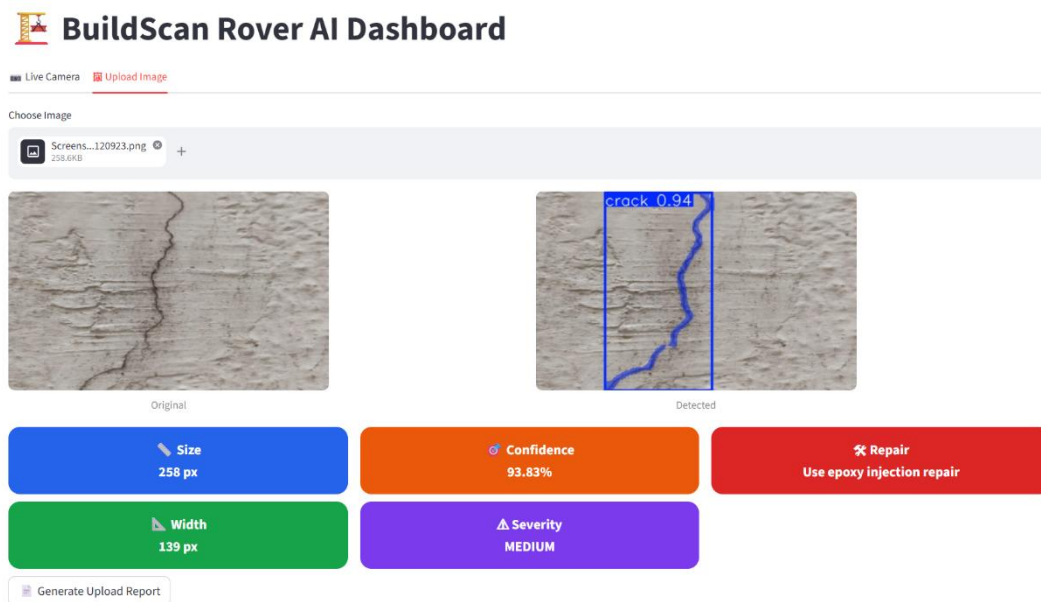


Figure 5.5 BuildScan Rover AI Dashboard – Tab-Based Multi-Mode Inspection

Fig 5.5 depicts the third dashboard page represents an improved tab-based interface containing both live camera mode and upload image mode. It allows flexible switching between real-time monitoring and static image analysis. The interface improves usability and provides a professional inspection environment.

5.4 AUTOMATIC PDF REPORT GENERATION

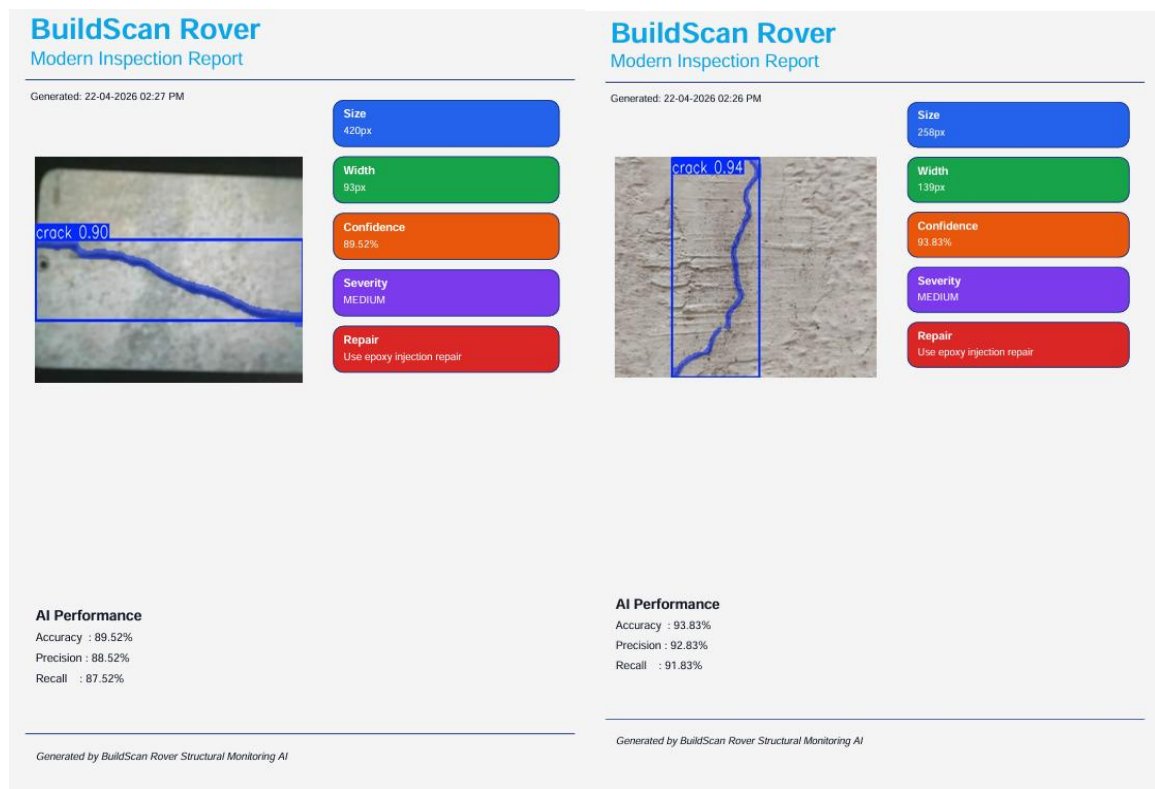


Figure 5.6 Automatically Generated PDF Inspection Report

Fig 5.6 depicts an automatic report generation feature was also implemented in the system. After crack detection, the dashboard generates an A4-size PDF inspection report containing the detected crack image, crack size, width, confidence score, severity level, repair suggestion, and AI performance metrics such as accuracy, precision, and recall.

The generated reports are time-stamped and can be downloaded instantly for documentation purposes. These reports help maintain structural inspection records and reduce manual paperwork. The PDF output demonstrates the practical usefulness of BuildScan Rover for civil maintenance, auditing, and preventive structural monitoring.

6. CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The BuildScan Rover project successfully developed a smart, portable, and cost-effective robotic system for indoor structural health inspection and crack monitoring. The rover integrates Arduino Uno, ESP32-CAM, ultrasonic sensor, Bluetooth communication, motor control modules, and the YOLO26n-seg AI model to perform real-time crack detection through a Streamlit dashboard. The system can move remotely, stream live video, inspect walls and narrow indoor areas, identify cracks, estimate crack size and severity, recommend suitable repair solutions, and generate automatic PDF reports for documentation. The developed prototype reduces manual inspection effort, improves worker safety, increases inspection speed, and provides accurate monitoring results. Overall, the project demonstrates the successful integration of robotics, IoT, and artificial intelligence to create an efficient solution for modern building maintenance, preventive inspection, and structural monitoring applications.

6.2 FUTURE SCOPE

Although the current system achieved the expected objectives, several improvements can be implemented in future versions to enhance performance and capabilities.

- Addition of fully autonomous navigation using SLAM and path planning algorithms.
- Integration of GPS for outdoor structural inspection applications.
- Use of higher resolution cameras for better crack image quality.
- Implementation of multi-defect detection such as corrosion, spalling, leakage, and surface damage.
- Cloud database integration for storing inspection history and remote monitoring.
- Mobile application development for direct control and monitoring.
- Real-time crack depth estimation using stereo vision or LiDAR sensors.
- Solar charging or improved battery backup for longer operation time.
- AI model optimization for faster inference on edge devices.
- Deployment in bridges, tunnels, dams, factories, and heritage buildings.

With these future enhancements, BuildScan Rover can be transformed into a fully intelligent autonomous structural inspection robot for large-scale industrial and civil monitoring applications.

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PUBLICATION

International Conference

1. Karthikeyan S, Yohendra S, Anandhan V, Vasanthakumar A , Ruthran S and Dr. P. Sudhakar, “BuildScan Rover: AI-Based Autonomous Robot for Indoor Structural Health Inspection”, International Conference on Intelligent Computing and Explainable AI (ICICEA’26), A.V.C. College of Engineering, Mannampandal. [ISBN: 978-93-5717-038-3] , April 2026